

EE115B *Digital Circuits*

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Part 3

Number Systems & Conversions

Number Systems

Digital signals are quantized ...

<u>Decimal</u>	<u>Binary</u>	<u>Octal</u>	<u>Hex</u>
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F



Decimal Numbers

Decimal numbers can be expressed as the sum of the products of each digit times the column value for that digit. Thus, the number 9240 can be expressed as

$$(9 \times 10^3) + (2 \times 10^2) + (4 \times 10^1) + (0 \times 10^0)$$

or

$$9 \times 1,000 + 2 \times 100 + 4 \times 10 + 0 \times 1$$

Example

Express the number 480.52 as the sum of values of each digit.

Solution

$$480.52 = (4 \times 10^2) + (8 \times 10^1) + (0 \times 10^0) + (5 \times 10^{-1}) + (2 \times 10^{-2})$$



Binary Numbers to Decimal

The Weighting Structure

$$2^{n-1} \dots 2^3 2^2 2^1 2^0 . 2^{-1} 2^{-2} \dots 2^{-n}$$

- With n bits one can count up to a number equal to $2^n - 1$
- 8-bit grouping is known as byte

Binary-to-Decimal Conversion

$$(1011001)_2 = (?)_{10}$$

$$(101.110)_2 = (?)_{10}$$



Hexadecimal Numbers (1)

Hexadecimal uses sixteen characters to represent numbers: the numbers 0 through 9 and the alphabetic characters A through F.

Large binary number can easily be converted to hexadecimal by grouping bits 4 at a time and writing the equivalent hexadecimal character.

Example

Express $1001\ 0110\ 0000\ 1110_2$ in hexadecimal:

Solution

Group the binary number by 4-bits starting from the right. Thus, **960E**

Decimal	Hexadecimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Hexadecimal Numbers (2)

Hexadecimal is a weighted number system. The column weights are powers of 16, which increase from right to left.

$$\text{Column weights } \begin{cases} 16^3 & 16^2 & 16^1 & 16^0 \\ 4096 & 256 & 16 & 1 \end{cases}$$

Example Express $1A2F_{16}$ in decimal.

Solution Start by writing the column weights:

4096	256	16	1
1	A	2	F_{16}

$$1(4096) + 10(256) + 2(16) + 15(1) = 6703_{10}$$

Decimal	Hexadecimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Octal Numbers (1)

Octal Numbers

Octal uses eight characters the numbers 0 through 7 to represent numbers.

There is no 8 or 9 character in octal.

Binary number can easily be converted to octal by grouping bits 3 at a time and writing the equivalent octal character for each group.

Example Express $1\ 001\ 011\ 000\ 001\ 110_2$ in octal:

Solution Group the binary number by 3-bits starting from the right. Thus, 113016_8

Decimal	Octal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	10	1000
9	11	1001
10	12	1010
11	13	1011
12	14	1100
13	15	1101
14	16	1110
15	17	1111



Octal Numbers (2)

Octal is also a weighted number system. The column weights are powers of 8, which increase from right to left.

$$\text{Column weights } \begin{cases} 8^3 & 8^2 & 8^1 & 8^0 \\ 512 & 64 & 8 & 1 \end{cases}$$

Example Express 3702_8 in decimal.

Solution Start by writing the column weights:

512	64	8	1
3	7	0	2_8

$$3(512) + 7(64) + 0(8) + 2(1) = 1986_{10}$$

Decimal	Octal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	10	1000
9	11	1001
10	12	1010
11	13	1011
12	14	1100
13	15	1101
14	16	1110
15	17	1111



Hex2Bin & Bin2Hex

- The hexadecimal number system has a base of sixteen. One can use “h” or subscript 16 to designate hexadecimal numbers.

Counting in Hexadecimal

..., E, F, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 1A, 1B, 1C, 1D, 1E, 1F, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 2A, 2B, 2C, 2D, 2E, 2F, 30, 31, ...

Binary-to-Hexadecimal Conversion

$$\begin{array}{ccccccc} 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ C & A & 5 & 7 & & & & & & & & & & \\ & & & & = & CA57_{16} & & & & & & & & \end{array}$$

Hexadecimal-to-Binary Conversion

$$\begin{array}{ccccccc} 1 & & 0 & & A & & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ & & & & & & & & & & & & \end{array}$$

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F



Oct2Bin & Bin2Oct

- The octal number system is composed of eight digits, 0, 1, 2, 3, 4, 5, 6, 7
- Counting octal number: 10, 11, 12, 13, 14, 15, 16, 17, 20, 21, ...
- an “o”, “Q” or subscript 8 are used to indicate an octal number.

Octal-to-Binary Conversion

$$\begin{array}{cccc} 7 & 5 & 2 & 6 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \underbrace{111} & \underbrace{101} & \underbrace{010} & \underbrace{110} \end{array}$$

Binary-to-Octal Conversion

$$\begin{array}{cccc} \underbrace{011} & \underbrace{010} & \underbrace{0000} & \underbrace{100} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 2 & 0 & 4 = 3204_8 \end{array}$$

Thought exercise: Oct2Hex & Hex2Oct?



Part 4

Number Conversions: (Advanced)

Conversions: Decimal to Binary (Integer)

Method 1: Try and Error (from high bits)

You can convert a decimal whole number to binary by reversing the procedure. Write the decimal weight of each column and place 1's in the columns that sum to the decimal number.

Example

Convert the decimal number 49 to binary.

Solution

The column weights double in each position to the right. Write down column weights until the last number is larger than the one you want to convert.

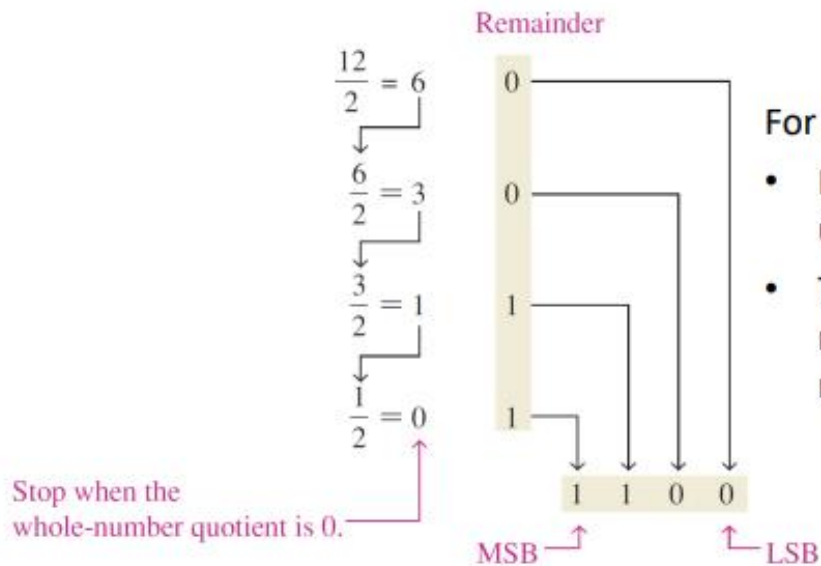
2^6	2^5	2^4	2^3	2^2	2^1	2^0	.
64	32	16	8	4	2	1	.
0	1	1	0	0	0	1	.



Conversions: Decimal to Binary (Integer)

Method 2: Division-by-2

Division-by-2



For example, to convert 12,

- Dividing 12 by 2, then divide each resulting quotient by 2 until a 0 quotient.
- The remainders generated by each division form the binary number. The first remainder is the LSB, and the last remainder is the MSB.

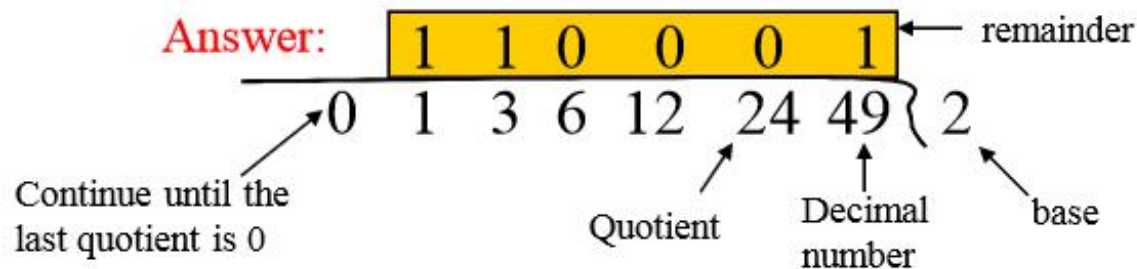


Conversions: Decimal to Other Base (Integer)

You can convert decimal to any other base by repeatedly dividing by the base. For binary, repeatedly divide by 2:

Example Convert the decimal number 49 to binary by repeatedly dividing by 2.

Solution You can do this by “reverse division” and the answer will read from left to right. Put quotients to the left and remainders on top.



Conversions: Decimal to Binary (Fraction)

You can convert a decimal fraction to binary by repeatedly multiplying the fractional results of successive multiplications by 2. The carries form the binary number.

Example

Convert the decimal fraction 0.188 to binary by repeatedly multiplying the fractional results by 2.

Solution

$0.188 \times 2 = 0.376$	carry = 0	MSB ↓
$0.376 \times 2 = 0.752$	carry = 0	
$0.752 \times 2 = 1.504$	carry = 1	
$0.504 \times 2 = 1.008$	carry = 1	
$0.008 \times 2 = 0.016$	carry = 0	

Answer = .00110 (for five significant digits)

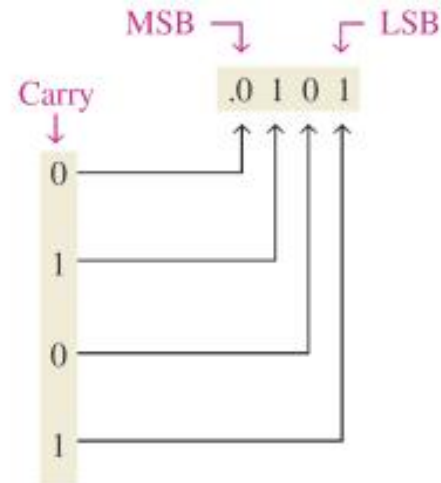


Conversions: Decimal to Binary (Fraction)

(One more example)

Multiplication-by-2

$$\begin{array}{l} 0.3125 \times 2 = 0.625 \\ \downarrow \\ 0.625 \times 2 = 1.25 \\ \downarrow \\ 0.25 \times 2 = 0.50 \\ \downarrow \\ 0.50 \times 2 = 1.00 \end{array}$$



Continue to the desired number of decimal places or stop when the fractional part is all zeros.

$$(21.375)_{10} = (?)_2$$

For example, to convert 0.3125,

- Begin by multiplying 0.3125 by 2 and then multiplying each resulting fractional part of the product by 2 until the fractional product is zero.
- The carry digits produce the binary number. The first carry produced is the MSB, and the last carry is the LSB



Thought Exercises

Other Bases to Decimal?

Other Bases to Binary?

Binary to Hex?

...

Python Conversions:

实例

```
binary_number = '101010'  
decimal_number = int(binary_number, 2) # 二进制转换为十进制  
octal_number = oct(decimal_number) # 十进制转换为八进制  
hexadecimal_number = hex(decimal_number) # 十进制转换为十六进制  
  
print('二进制数:', binary_number)  
print('转换为十进制:', decimal_number)  
print('转换为八进制:', octal_number)  
print('转换为十六进制:', hexadecimal_number)
```

<https://www.runoob.com/python3/python3-conversion-binary-octal-hexadecimal.html>



Thought Exercises

Solve these questions through manual calculation:

- Convert the binary number 101101 to decimal, hexadecimal, and octal.
- Convert the decimal number 52 to binary, hexadecimal, and octal.
- Convert the hexadecimal number A2F to binary and octal.
- Convert the octal number 637 to binary and decimal.
- Convert the decimal number 255 to binary and hexadecimal.
- Convert decimal fraction 0.188 to binary.